

Other outcomes: Ordinal, nominal, count, duration

Francisco Villamil

Applied Quantitative Methods II
MA in Social Sciences, Spring 2026

Logistics

- Presentations (April 16th)
 - **5-minute presentations**, blocks of ≈ 5 , random order, 15-min feedback after each block
<https://franvillamil.github.io/AQM2/logistics.html>
 - Send me **presentations** by email **in advance**
3 slides max!: e.g. what, how, problems
- Rest of the course
 - Computing session (April 23rd)
 - Final session (April 30th), **exam** + review

Today's goals

- Understand the GLM framework as a unifying structure
- Model ordinal outcomes with ordered logit
- Model nominal (multi-category) outcomes with multinomial logit
- Model count data with Poisson and negative binomial regression
- Model time-to-event data with survival analysis and the Cox model
- Know which model to reach for and why

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor

$$\eta_i = \mathbf{x}_i\boldsymbol{\beta}$$

A weighted sum of
covariates

Link function

$$g(\mu_i) = \eta_i$$

Connects mean to
predictor

Distribution

$$Y_i \sim F(\mu_i)$$

Data-generating process

- OLS is a GLM: identity link + normal distribution

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor

$$\eta_i = \mathbf{x}_i \boldsymbol{\beta}$$

A weighted sum of
covariates

Link function

$$g(\mu_i) = \eta_i$$

Connects mean to
predictor

Distribution

$$Y_i \sim F(\mu_i)$$

Data-generating process

- OLS is a GLM: identity link + normal distribution
- Logit is a GLM: logit link + Bernoulli distribution

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor

$$\eta_i = \mathbf{x}_i \boldsymbol{\beta}$$

A weighted sum of
covariates

Link function

$$g(\mu_i) = \eta_i$$

Connects mean to
predictor

Distribution

$$Y_i \sim F(\mu_i)$$

Data-generating process

- OLS is a GLM: identity link + normal distribution
- Logit is a GLM: logit link + Bernoulli distribution
- Choosing a model = choosing a link + distribution

Matching outcome type to model

Outcome type	Distribution	Link	Model / Function
Continuous	Normal	Identity	OLS (<code>lm()</code>)
Binary (0/1)	Bernoulli	Logit	Logit (<code>glm(...,binomial)</code>)
Ordinal	Prop. odds	Cumul. logit	Ordered logit (<code>polr()</code>)
Nominal (>2)	Multinomial	Log-odds	Multinom. logit (<code>multinom()</code>)
Count (≥ 0)	Poisson	Log	Poisson (<code>glm(...,poisson)</code>)
Count (overdisp)	Neg. Binomial	Log	NegBin (<code>glm.nb()</code>)
Duration (time)	—	—	Cox PH (<code>coxph()</code>)

First question: what is the data-generating process for your outcome?

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Ordinal outcomes:

When categories have an order but not equal spacing

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
 - Survey: “How satisfied are you?” (1 = not at all, 5 = very)

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
 - Survey: “How satisfied are you?” (1 = not at all, 5 = very)
 - Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
 - Survey: “How satisfied are you?” (1 = not at all, 5 = very)
 - Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
 - Democracy rating: Autocracy → Hybrid → Democracy

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
 - Survey: “How satisfied are you?” (1 = not at all, 5 = very)
 - Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
 - Democracy rating: Autocracy → Hybrid → Democracy
 - Conflict intensity: No conflict → Low → Medium → High

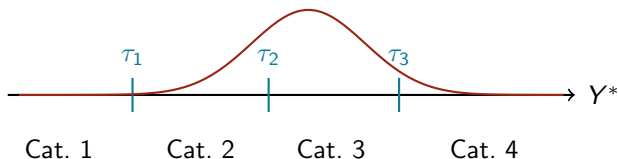
Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
 - Survey: “How satisfied are you?” (1 = not at all, 5 = very)
 - Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
 - Democracy rating: Autocracy → Hybrid → Democracy
 - Conflict intensity: No conflict → Low → Medium → High
- The distance between “1” and “2” need not equal the distance between “2” and “3”

Ordinal outcomes are ordered categories

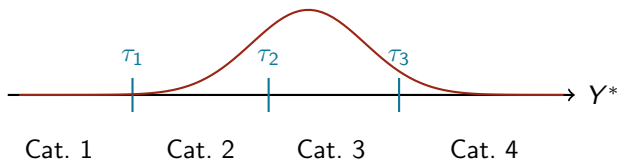
- Categories have a natural **ordering**, but not equal intervals
- Examples:
 - Survey: “How satisfied are you?” (1 = not at all, 5 = very)
 - Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
 - Democracy rating: Autocracy → Hybrid → Democracy
 - Conflict intensity: No conflict → Low → Medium → High
- The distance between “1” and “2” need not equal the distance between “2” and “3”
- OLS treats them as equally spaced — often wrong

The latent variable idea



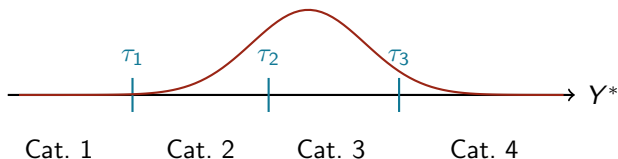
- Y^* : unobserved continuous variable (e.g., “true” satisfaction)

The latent variable idea



- Y^* : unobserved continuous variable (e.g., “true” satisfaction)
- We observe category j when $\tau_{j-1} < Y^* \leq \tau_j$

The latent variable idea



- Y^* : unobserved continuous variable (e.g., “true” satisfaction)
- We observe category j when $\tau_{j-1} < Y^* \leq \tau_j$
- Thresholds τ_j are **estimated** from the data

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i\beta))}$$

- One set of β for all thresholds (**proportional odds assumption**)

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i\beta))}$$

- One set of β for all thresholds (**proportional odds assumption**)
- $\beta_k > 0$ shifts Y^* upward $\rightarrow$ **higher** categories become more likely

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i\beta))}$$

- One set of β for all thresholds (**proportional odds assumption**)
- $\beta_k > 0$ shifts Y^* upward $\rightarrow$ **higher** categories become more likely
- $\Pr(Y_i = j) = \Pr(Y_i \leq j) - \Pr(Y_i \leq j - 1)$: probability for each category by differencing cumulative probabilities

Ordered logit in R

```
library(MASS); library(marginaleffects)
m_ord = polr(factor(satisfaction) ~ age + educ,
              data = mydata, Hess = TRUE)
avg_slopes(m_ord) # AMEs per predictor per category
```

- Hess = TRUE: stores the Hessian — required for SEs

Ordered logit in R

```
library(MASS); library(marginaleffects)
m_ord = polr(factor(satisfaction) ~ age + educ,
             data = mydata, Hess = TRUE)
avg_slopes(m_ord) # AMEs per predictor per category
```

- Hess = TRUE: stores the Hessian — required for SEs
- **Sign warning:** raw polr() coefficients use a reversed sign convention — always interpret via avg_slopes()

Ordered logit: R output

Using housing data from MASS: satisfaction with housing conditions

```
> m_ord = polr(Sat ~ Infl + Type + Cont, weights = Freq, data = housing, Hess = TRUE)
> summary(m_ord)
```

Coefficients:

	Value	Std. Error	t value
InflMedium	0.5664	0.10465	5.412
InflHigh	1.2888	0.12716	10.136
TypeApartment	-0.5724	0.11924	-4.800
TypeAtrium	-0.3662	0.15517	-2.360
TypeTerrace	-1.0910	0.15149	-7.202
ContHigh	0.3603	0.09554	3.771

Intercepts:

	Value	Std. Error	t value
Low Medium	-0.4961	0.1248	-3.9739
Medium High	0.6907	0.1255	5.5049

Residual Deviance: 3479.149

AIC: 3495.149

- **Coefficients:** log-odds scale, not directly interpretable
- **Intercepts:** estimated thresholds τ_j between categories
- No p -values — use t -values > 2 as rough guide

Ordered logit: marginal effects

avg_slopes(m_ord) — average marginal effects on probability scale

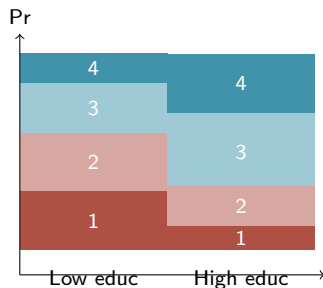
```
> avg_slopes(m_ord)
```

Term	Group	Contrast	Estimate	Std. Error	z	Pr(> z)
Cont	Low	High - Low	-0.0727	0.0193	-3.77	<0.001
Cont	Medium	High - Low	-0.0064	0.0022	-2.93	0.003
Cont	High	High - Low	0.0791	0.0206	3.83	<0.001
Infl	Low	Medium - Low	-0.1287	0.0236	-5.46	<0.001
Infl	Low	High - Low	-0.2619	0.0237	-11.04	<0.001
Infl	High	Medium - Low	0.1202	0.0220	5.46	<0.001
Infl	High	High - Low	0.2909	0.0279	10.42	<0.001
Type	Low	Terrace - Tower	0.2239	0.0312	7.17	<0.001
Type	High	Terrace - Tower	-0.2393	0.0316	-7.59	<0.001

- One row per predictor $\times$ outcome category
- **Group**: which satisfaction level (Low, Medium, High)
- E.g., high influence $\rightarrow$ +29pp probability of “High” satisfaction
- AMEs across categories **sum to zero** for each predictor

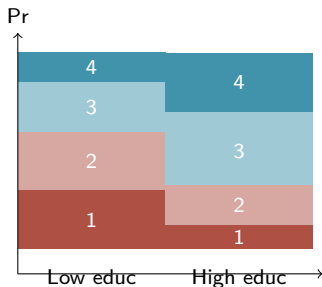
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive



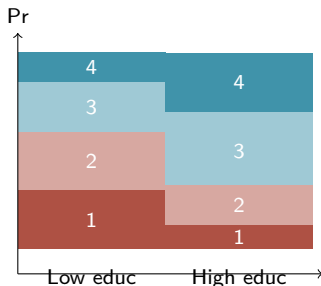
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects



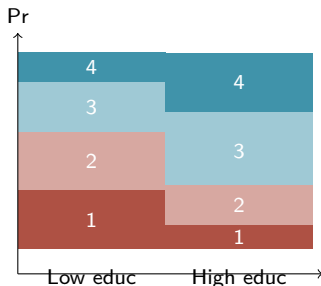
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better:** average marginal effects
 - One AME per predictor *per category*



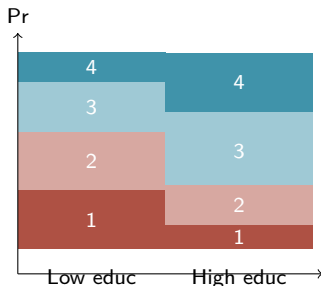
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better:** average marginal effects
 - One AME per predictor *per category*
 - “A one-unit increase in education changes the probability of category j by ΔP_j ”



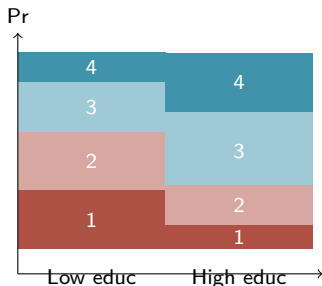
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better:** average marginal effects
 - One AME per predictor *per category*
 - “A one-unit increase in education changes the probability of category j by ΔP_j ”
- **Best:** predicted probability plots



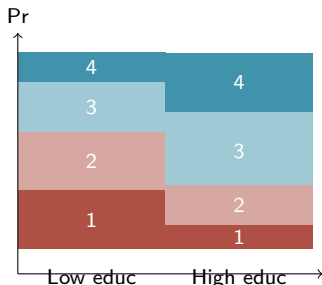
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better:** average marginal effects
 - One AME per predictor *per category*
 - “A one-unit increase in education changes the probability of category j by ΔP_j ”
- **Best:** predicted probability plots
 - How does $\Pr(Y = j)$ change across X ?



Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better:** average marginal effects
 - One AME per predictor *per category*
 - “A one-unit increase in education changes the probability of category j by ΔP_j ”
- **Best:** predicted probability plots
 - How does $\Pr(Y = j)$ change across X ?
 - Use `plot_predictions(m_ord, condition = "educ")`



Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit
- The ordinal structure constrains transitions to respect the natural ordering — large jumps (e.g., autocracy → full democracy) are possible but predicted to be rare

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit
- The ordinal structure constrains transitions to respect the natural ordering — large jumps (e.g., autocracy → full democracy) are possible but predicted to be rare
- Covariates shift the latent propensity, making upward or downward transitions more likely

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit
- The ordinal structure constrains transitions to respect the natural ordering — large jumps (e.g., autocracy → full democracy) are possible but predicted to be rare
- Covariates shift the latent propensity, making upward or downward transitions more likely
 - E.g., economic crisis → higher probability of moving down the scale

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit
- The ordinal structure constrains transitions to respect the natural ordering — large jumps (e.g., autocracy → full democracy) are possible but predicted to be rare
- Covariates shift the latent propensity, making upward or downward transitions more likely
 - E.g., economic crisis → higher probability of moving down the scale

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit
- The ordinal structure constrains transitions to respect the natural ordering — large jumps (e.g., autocracy → full democracy) are possible but predicted to be rare
- Covariates shift the latent propensity, making upward or downward transitions more likely
 - E.g., economic crisis → higher probability of moving down the scale

Application: Markov transition models

- Ordinal models can be used to study **transitions between ordered states**
 - Democracy scores (autocracy → anocracy → democracy)
 - Credit ratings (AAA → AA → A → ...)
 - Health status (healthy → mild → severe → critical)
- **Idea:** given current state S_t , model the probability of transitioning to each state at S_{t+1} using ordinal logit
- The ordinal structure constrains transitions to respect the natural ordering — large jumps (e.g., autocracy → full democracy) are possible but predicted to be rare
- Covariates shift the latent propensity, making upward or downward transitions more likely
 - E.g., economic crisis → higher probability of moving down the scale

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Nominal outcomes:

Multiple unordered categories — vote choice, transport mode,
conflict type

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:
 - Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, ...)

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:
 - Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, ...)
 - Mode of transport (car, bus, train, bike, walk)

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:
 - Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, ...)
 - Mode of transport (car, bus, train, bike, walk)
 - Type of conflict (interstate, civil war, non-state)

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:
 - Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, ...)
 - Mode of transport (car, bus, train, bike, walk)
 - Type of conflict (interstate, civil war, non-state)
 - Occupation category (manager, professional, manual, service)

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:
 - Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, ...)
 - Mode of transport (car, bus, train, bike, walk)
 - Type of conflict (interstate, civil war, non-state)
 - Occupation category (manager, professional, manual, service)
- Ordered logit is wrong here — it would impose an ordering

When categories have no natural order

- **Nominal outcomes:** categories, but no ranking
- Examples:
 - Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, ...)
 - Mode of transport (car, bus, train, bike, walk)
 - Type of conflict (interstate, civil war, non-state)
 - Occupation category (manager, professional, manual, service)
- Ordered logit is wrong here — it would impose an ordering
- We need a model that treats all categories symmetrically

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\beta'_j \mathbf{x}_i)}{\sum_{k=1}^J \exp(\beta'_k \mathbf{x}_i)}$$

- J categories $\rightarrow J - 1$ sets of coefficients (one category is the **reference**)

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\beta'_j \mathbf{x}_i)}{\sum_{k=1}^J \exp(\beta'_k \mathbf{x}_i)}$$

- J categories $\rightarrow J - 1$ sets of coefficients (one category is the **reference**)
- Reference category: $\beta_1 = \mathbf{0}$ by convention

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\beta'_j \mathbf{x}_i)}{\sum_{k=1}^J \exp(\beta'_k \mathbf{x}_i)}$$

- J categories $\rightarrow J - 1$ sets of coefficients (one category is the **reference**)
- Reference category: $\beta_1 = \mathbf{0}$ by convention
- Each β_{jk} : effect of x_k on log-odds of category j **vs. reference**

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\beta'_j \mathbf{x}_i)}{\sum_{k=1}^J \exp(\beta'_k \mathbf{x}_i)}$$

- J categories $\rightarrow J - 1$ sets of coefficients (one category is the **reference**)
- Reference category: $\beta_1 = \mathbf{0}$ by convention
- Each β_{jk} : effect of x_k on log-odds of category j **vs. reference**
- All predicted probabilities sum to 1: $\sum_{j=1}^J \Pr(Y_i = j) = 1$

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
                  data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
```

- First category = reference; coefficients = log-odds vs. reference

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
                  data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
```

- First category = reference; coefficients = log-odds vs. reference
- Change reference: `relevel(factor(vote), ref = "Labour")`

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
                  data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
```

- First category = reference; coefficients = log-odds vs. reference
- Change reference: `relevel(factor(vote), ref = "Labour")`
- Interpret via `avg_slopes()` or `plot_predictions()`

Multinomial logit: R output

Using iris data: predicting species from flower measurements

```
> m_mnom = multinom(Species ~ Sepal.Length + Sepal.Width + Petal.Length,  
+                   data = iris, trace = FALSE)  
> summary(m_mnom)
```

Coefficients:

	(Intercept)	Sepal.Length	Sepal.Width	Petal.Length
versicolor	15.84477	-5.106783	-9.373084	15.67647
virginica	-22.68156	-8.980237	-9.993958	28.90429

Std. Errors:

	(Intercept)	Sepal.Length	Sepal.Width	Petal.Length
versicolor	38.86805	104.8927	164.5751	79.52894
virginica	39.08951	104.9167	164.5911	79.55610

Residual Deviance: 23.77288

AIC: 39.77288

- One row of coefficients **per non-reference category**
- Reference = *setosa*; each row = log-odds vs. reference
- Coefficients are log-odds → hard to interpret directly
- Use `avg_slopes()` or `plot_predictions()` instead

Multinomial logit: marginal effects

`avg_slopes(m_mnom)` — average marginal effects on probability scale

```
> avg_slopes(m_mnom)
```

Term	Group	Estimate	Std. Error	z	Pr(> z)
Petal.Length	setosa	-0.00003018	0.001756	-0.017	0.9863
Petal.Length	versicolor	-0.31811938	0.023102	-13.770	<0.001
Petal.Length	virginica	0.31814956	0.023035	13.812	<0.001
Sepal.Length	setosa	0.00000983	0.000578	0.017	0.9864
Sepal.Length	versicolor	0.09315278	0.032003	2.911	0.0036
Sepal.Length	virginica	-0.09316261	0.031998	-2.912	0.0036
Sepal.Width	setosa	0.00001805	0.001106	0.016	0.9870
Sepal.Width	versicolor	0.01491495	0.054905	0.272	0.7859
Sepal.Width	virginica	-0.01493299	0.054891	-0.272	0.7856

- One row per predictor $\times$ category (like ordered logit)
- 1cm longer petal $\rightarrow -32\text{pp}$ *versicolor*, $+32\text{pp}$ *virginica*
- AMEs across categories **sum to zero** for each predictor

The IIA assumption

Independence of Irrelevant Alternatives (IIA): odds ratio btw any two alternatives **unaffected** by presence of other alternatives

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

- The famous **red bus / blue bus** problem:

The IIA assumption

Independence of Irrelevant Alternatives (IIA): odds ratio btw any two alternatives **unaffected** by presence of other alternatives

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

- The famous **red bus / blue bus** problem:
 - 50% car, 50% red bus — now introduce a blue bus

The IIA assumption

Independence of Irrelevant Alternatives (IIA): odds ratio btw any two alternatives **unaffected** by presence of other alternatives

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

- The famous **red bus / blue bus** problem:
 - 50% car, 50% red bus — now introduce a blue bus
 - IIA implies: 33% car, 33% red bus, 33% blue bus

The IIA assumption

Independence of Irrelevant Alternatives (IIA): odds ratio btw any two alternatives **unaffected** by presence of other alternatives

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

- The famous **red bus / blue bus** problem:
 - 50% car, 50% red bus — now introduce a blue bus
 - IIA implies: 33% car, 33% red bus, 33% blue bus
 - Reality: 50% car, 25% red bus, 25% blue bus

The IIA assumption

Independence of Irrelevant Alternatives (IIA): odds ratio btw any two alternatives **unaffected** by presence of other alternatives

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

- The famous **red bus / blue bus** problem:
 - 50% car, 50% red bus — now introduce a blue bus
 - IIA implies: 33% car, 33% red bus, 33% blue bus
 - Reality: 50% car, 25% red bus, 25% blue bus
- IIA holds when alternatives are **genuinely distinct**

The IIA assumption

Independence of Irrelevant Alternatives (IIA): odds ratio btw any two alternatives **unaffected** by presence of other alternatives

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

- The famous **red bus / blue bus** problem:
 - 50% car, 50% red bus — now introduce a blue bus
 - IIA implies: 33% car, 33% red bus, 33% blue bus
 - Reality: 50% car, 25% red bus, 25% blue bus
- IIA holds when alternatives are **genuinely distinct**
- Violations: similar alternatives that substitute for each other

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

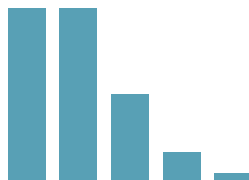
Model Selection in Practice

Count data:

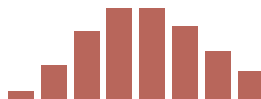
Non-negative integers — conflicts, protests, bills, articles

Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...



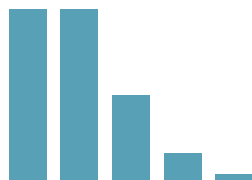
$\lambda = 1$



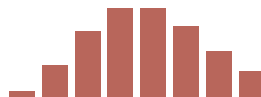
$\lambda = 4$

Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks



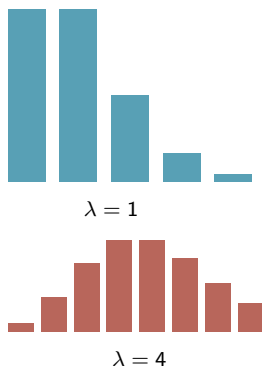
$\lambda = 1$



$\lambda = 4$

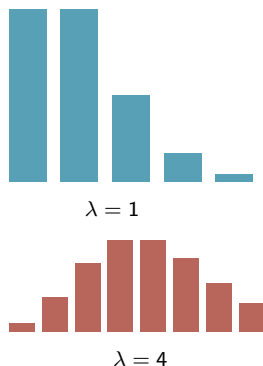
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**



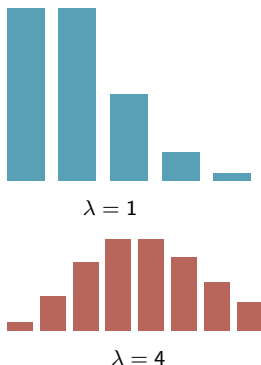
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**
 - Counts ≥ 0 : OLS can predict negative values



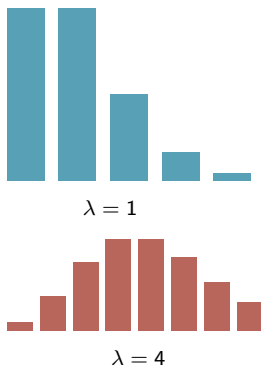
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**
 - Counts ≥ 0 : OLS can predict negative values
 - Variance increases with the mean



Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**
 - Counts ≥ 0 : OLS can predict negative values
 - Variance increases with the mean
 - Strongly right-skewed; often many zeros



The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i\boldsymbol{\beta})$$

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i\boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i\boldsymbol{\beta})$$

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i\boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i\boldsymbol{\beta})$$

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i\boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative
- Coefficient interpretation: $\exp(\beta_k) = \mathbf{multiplicative}$ change in expected count

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i\boldsymbol{\beta})$$

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i\boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative
- Coefficient interpretation: $\exp(\beta_k) = \mathbf{multiplicative}$ change in expected count
 - $\beta_k = 0.3 \Rightarrow \exp(0.3) \approx 1.35$: 35% increase in expected count

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i\boldsymbol{\beta})$$

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i\boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative
- Coefficient interpretation: $\exp(\beta_k) = \mathbf{multiplicative}$ change in expected count
→ $\beta_k = 0.3 \Rightarrow \exp(0.3) \approx 1.35$: 35% increase in expected count
- Poisson assumption: $\text{Var}(Y_i) = E[Y_i]$ (mean = variance)

Poisson in R and interpretation

```
# Fit Poisson model  
m_pois = glm(n_conflicts ~ gdp_pc + pop_l,  
             family = "poisson", data = mydata)  
  
summary(m_pois)  
exp(coef(m_pois)) # incidence rate ratios
```

Raw coefficient

$$\hat{\beta}_{\text{gdp_pc}} = -0.24$$

Hard to interpret directly

Exponentiated

$$\exp(-0.24) \approx 0.79$$

→ 21% fewer conflicts per unit GDP

Also: `avg_slopes(m_pois)` gives AMEs on count scale

Poisson: R output

Using warpbreaks: number of breaks in yarn by wool type and tension

```
> m_pois = glm(breaks ~ wool + tension, family = "poisson", data = warpbreaks)
> summary(m_pois)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.69196	0.04541	81.302	< 2e-16 ***
woolB	-0.20599	0.05157	-3.994	6.49e-05 ***
tensionM	-0.32132	0.06027	-5.332	9.73e-08 ***
tensionH	-0.51849	0.06396	-8.107	5.21e-16 ***

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 297.37 on 53 degrees of freedom
Residual deviance: 210.39 on 50 degrees of freedom
AIC: 493.06

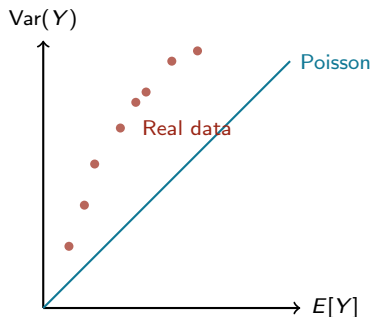
```
> exp(coef(m_pois))
```

(Intercept)	woolB	tensionM	tensionH
40.124	0.814	0.725	0.595

- Coefficients on **log scale** → exponentiate for IRRs
- woolB: $\exp(-0.21) = 0.81 \rightarrow 19\%$ fewer breaks
- **Red flag:** residual deviance (210) $\gg$ df (50) → overdispersion!

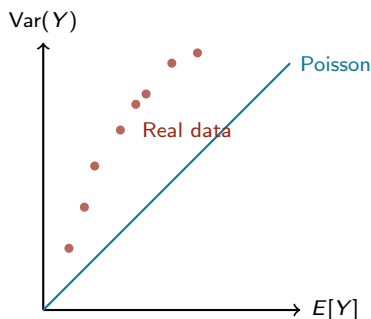
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$



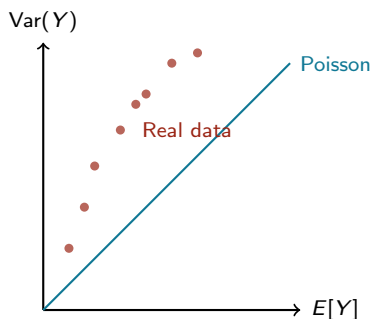
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always:
 $\text{Var}(Y) \gg E[Y]$



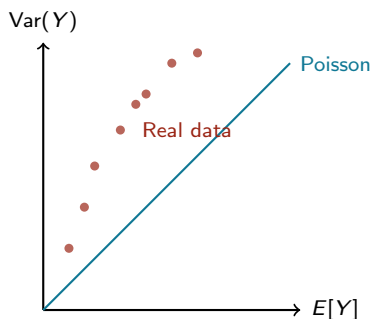
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always:
 $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:



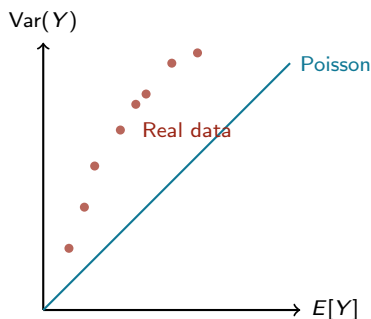
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always:
 $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
 - Standard errors too **small**



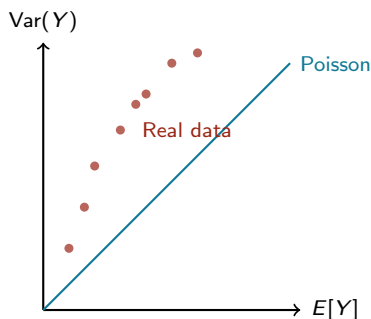
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always:
 $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
 - Standard errors too **small**
 - p -values too **small**



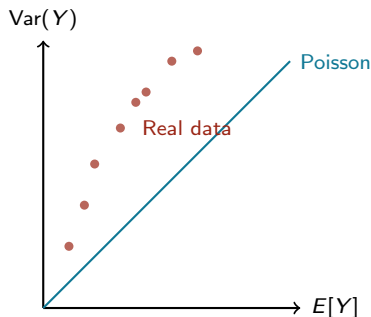
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always:
 $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
 - Standard errors too **small**
 - p -values too **small**
 - False positives



The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always:
 $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
 - Standard errors too **small**
 - p -values too **small**
 - False positives
- Quick check:
Residual deviance $\gg$ df



Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
→ $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
 - $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)
 - Small θ : high overdispersion

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
 - $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)
 - Small θ : high overdispersion
- Same interpretation as Poisson: $\exp(\beta_k) = \text{incidence rate ratio}$

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
 - $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)
 - Small θ : high overdispersion
- Same interpretation as Poisson: $\exp(\beta_k) = \text{incidence rate ratio}$
- In R: `MASS::glm.nb(y ~ x1 + x2)`

Negative binomial: R output

Same warpbreaks data — compare with Poisson output

```
> m_nb = glm.nb(breaks ~ wool + tension, data = warpbreaks)
> summary(m_nb)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.6734	0.0979	37.520	< 2e-16 ***
woolB	-0.1862	0.1010	-1.844	0.0651 .
tensionM	-0.2992	0.1217	-2.458	0.0140 *
tensionH	-0.5114	0.1237	-4.133	3.58e-05 ***

(Dispersion parameter for Negative Binomial(9.9444) family taken to be 1)

Null deviance: 75.464 on 53 degrees of freedom
Residual deviance: 53.723 on 50 degrees of freedom
AIC: 408.76

Theta: 9.94
Std. Err.: 2.56

```
> exp(coef(m_nb))
(Intercept)      woolB      tensionM      tensionH
    39.384      0.830      0.741      0.600
```

- **SEs are larger** than Poisson (e.g., woolB: 0.10 vs. 0.05)
- woolB now only marginal ($p = 0.065$) — was “highly significant”
- Deviance/df ≈ 1.07 — overdispersion is handled

Count models: a summary

Model		R function	When to use
Poisson		<code>glm(...,poisson)</code>	Equidispersed counts
Negative binomial		<code>MASS::glm.nb()</code>	Overdispersed counts
Zero-inflated	Pois-	<code>pscl::zeroinfl()</code>	Excess zeros, equidisp.
son			
Zero-inflated	Neg-	<code>pscl::zeroinfl()</code>	Excess zeros, overdisp.
Bin			

- **Default:** start Poisson, test overdispersion, switch to NegBin if needed (**in practice:** often go ahead to NB)

Count models: a summary

Model		R function	When to use
Poisson		<code>glm(...,poisson)</code>	Equidispersed counts
Negative binomial		<code>MASS::glm.nb()</code>	Overdispersed counts
Zero-inflated	Pois-	<code>pscl::zeroinfl()</code>	Excess zeros, equidisp.
son			
Zero-inflated	Neg-	<code>pscl::zeroinfl()</code>	Excess zeros, overdisp.
Bin			

- **Default:** start Poisson, test overdispersion, switch to NegBin if needed (**in practice:** often go ahead to NB)
- Zero-inflation: two-process model (structural vs. count zeros)

Count models: a summary

Model		R function	When to use
Poisson		<code>glm(...,poisson)</code>	Equidispersed counts
Negative binomial		<code>MASS::glm.nb()</code>	Overdispersed counts
Zero-inflated	Pois-	<code>pscl::zeroinfl()</code>	Excess zeros, equidisp.
son			
Zero-inflated	Neg-	<code>pscl::zeroinfl()</code>	Excess zeros, overdisp.
Bin			

- **Default:** start Poisson, test overdispersion, switch to NegBin if needed (**in practice:** often go ahead to NB)
- Zero-inflation: two-process model (structural vs. count zeros)
- All share the log link: $\exp(\hat{\beta}) = \text{incidence rate ratio (IRR)}$

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**
 - Process 1 (logit): is the unit “at risk” of a positive count?

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**
 - Process 1 (logit): is the unit “at risk” of a positive count?
 - Process 2 (Poisson/NegBin): if at risk, how many events?

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**
 - Process 1 (logit): is the unit “at risk” of a positive count?
 - Process 2 (Poisson/NegBin): if at risk, how many events?
- Distinguishes **structural zeros** (unit can't have the event) from **sampling zeros** (could happen, just didn't)

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**
 - Process 1 (logit): is the unit “at risk” of a positive count?
 - Process 2 (Poisson/NegBin): if at risk, how many events?
- Distinguishes **structural zeros** (unit can't have the event) from **sampling zeros** (could happen, just didn't)
- In R: `pscl::zeroinfl(y ~ x | z, dist = "negbin")`

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**
 - Process 1 (logit): is the unit “at risk” of a positive count?
 - Process 2 (Poisson/NegBin): if at risk, how many events?
- Distinguishes **structural zeros** (unit can't have the event) from **sampling zeros** (could happen, just didn't)
- In R: `pscl::zeroinfl(y ~ x | z, dist = "negbin")`
 - x: predictors for the count process

Zero-inflated models

- Sometimes data have **more zeros than any count distribution predicts**
 - Conflict events: many country-years at peace (structural zeros)
 - Patent counts: many firms never innovate
- **Two-process model:**
 - Process 1 (logit): is the unit “at risk” of a positive count?
 - Process 2 (Poisson/NegBin): if at risk, how many events?
- Distinguishes **structural zeros** (unit can't have the event) from **sampling zeros** (could happen, just didn't)
- In R: `pscl::zeroinfl(y ~ x | z, dist = "negbin")`
 - x: predictors for the count process
 - z: predictors for the zero-inflation process

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Duration analysis:

Time until an event — war onset, cabinet survival, regime
collapse

Time-to-event outcomes

- Outcomes that record **how long until something happens**

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office
 - Time to re-arrest after prison release

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office
 - Time to re-arrest after prison release
 - How long an authoritarian regime survives

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office
 - Time to re-arrest after prison release
 - How long an authoritarian regime survives
 - Time until a bill is passed (from introduction)

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office
 - Time to re-arrest after prison release
 - How long an authoritarian regime survives
 - Time until a bill is passed (from introduction)
- Why not OLS on duration?

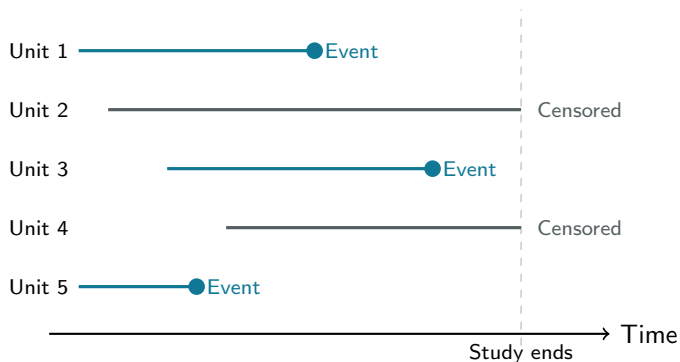
Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office
 - Time to re-arrest after prison release
 - How long an authoritarian regime survives
 - Time until a bill is passed (from introduction)
- Why not OLS on duration?
 - Durations are ≥ 0 and right-skewed

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
 - Time until a war begins (from peace)
 - Duration of a coalition government in office
 - Time to re-arrest after prison release
 - How long an authoritarian regime survives
 - Time until a bill is passed (from introduction)
- Why not OLS on duration?
 - Durations are ≥ 0 and right-skewed
 - **Censoring**: many observations end without the event occurring

The censoring problem



Censored units: event did not occur during observation window

What does survival data look like?

Unit	Spell start	Duration	Event	Covariates ...
Country A	1975	12 years	1 (breakdown)	GDP, region, ...
Country B	1991	30 years	0 (censored)	GDP, region, ...
Country C	1985	8 years	1 (breakdown)	GDP, region, ...
Country D	2002	19 years	0 (censored)	GDP, region, ...
Country E	1960	3 years	1 (breakdown)	GDP, region, ...

- Each row is a **spell**: one unit observed from entry to event or censoring

What does survival data look like?

Unit	Spell start	Duration	Event	Covariates ...
Country A	1975	12 years	1 (breakdown)	GDP, region, ...
Country B	1991	30 years	0 (censored)	GDP, region, ...
Country C	1985	8 years	1 (breakdown)	GDP, region, ...
Country D	2002	19 years	0 (censored)	GDP, region, ...
Country E	1960	3 years	1 (breakdown)	GDP, region, ...

- Each row is a **spell**: one unit observed from entry to event or censoring
- Two outcome variables: **duration** (T) and **event indicator** (d : 1 = event, 0 = censored)

What does survival data look like?

Unit	Spell start	Duration	Event	Covariates ...
Country A	1975	12 years	1 (breakdown)	GDP, region, ...
Country B	1991	30 years	0 (censored)	GDP, region, ...
Country C	1985	8 years	1 (breakdown)	GDP, region, ...
Country D	2002	19 years	0 (censored)	GDP, region, ...
Country E	1960	3 years	1 (breakdown)	GDP, region, ...

- Each row is a **spell**: one unit observed from entry to event or censoring
- Two outcome variables: **duration** (T) and **event indicator** (d : 1 = event, 0 = censored)
- In R: combined into a `Surv(time, event)` object

Key concepts: survival and hazard

Survival function $S(t)$

$$S(t) = \Pr(T > t)$$

Probability of surviving past time t

$$S(0) = 1, S(\infty) = 0$$

Monotonically non-increasing

- High $h(t)$: event is likely soon

Hazard function $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t ,
conditional on survival to t

“Danger per unit time at time t ”

Key concepts: survival and hazard

Survival function $S(t)$

$$S(t) = \Pr(T > t)$$

Probability of surviving past time t

$$S(0) = 1, S(\infty) = 0$$

Monotonically non-increasing

Hazard function $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t ,
conditional on survival to t

“Danger per unit time at time t ”

- High $h(t)$: event is likely soon
- $S(t)$ and $h(t)$ are two views of the same object

Key concepts: survival and hazard

Survival function $S(t)$

$$S(t) = \Pr(T > t)$$

Probability of surviving past time t

$$S(0) = 1, S(\infty) = 0$$

Monotonically non-increasing

Hazard function $h(t)$

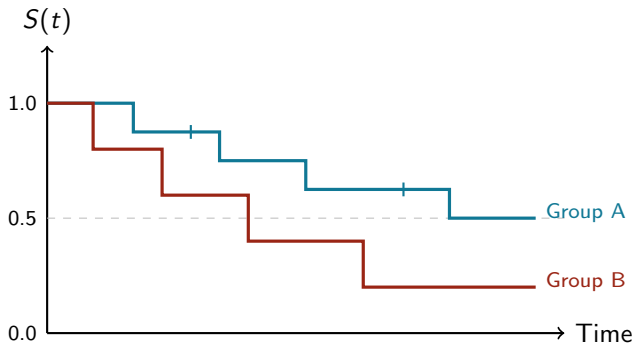
$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t ,
conditional on survival to t

“Danger per unit time at time t ”

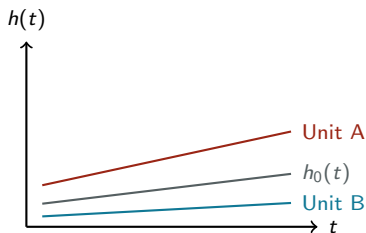
- High $h(t)$: event is likely soon
- $S(t)$ and $h(t)$ are two views of the same object
- Covariates shift $h(t)$ — that is what regression models

Kaplan-Meier: visualizing survival



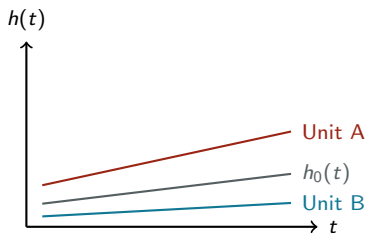
Kaplan-Meier estimator: non-parametric, no covariates, visual comparison of groups

The proportional hazards idea



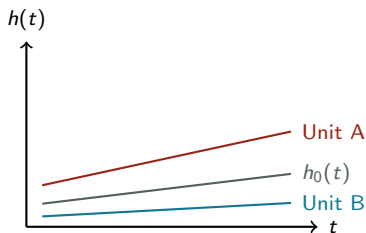
- **Key idea:** each unit has its own hazard, but all are proportional shifts of the same baseline $h_0(t)$

The proportional hazards idea



- **Key idea:** each unit has its own hazard, but all are proportional shifts of the same baseline $h_0(t)$
- The ratio $h_i(t)/h_j(t)$ is **constant over time** — curves never cross

The proportional hazards idea



- **Key idea:** each unit has its own hazard, but all are proportional shifts of the same baseline $h_0(t)$
- The ratio $h_i(t)/h_j(t)$ is **constant over time** — curves never cross
- Violation: if the gap between lines closes or reverses over time $\Rightarrow$ PH assumption fails

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i\boldsymbol{\beta})$: multiplicative shift of the baseline

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i\boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i\boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*
- $\exp(\beta_k)$: **hazard ratio** for a one-unit increase in x_k

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i\boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*
- $\exp(\beta_k)$: **hazard ratio** for a one-unit increase in x_k
 - $\exp(\beta_k) > 1$: higher hazard (event happens sooner)

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i\boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*
- $\exp(\beta_k)$: **hazard ratio** for a one-unit increase in x_k
 - $\exp(\beta_k) > 1$: higher hazard (event happens sooner)
 - $\exp(\beta_k) < 1$: lower hazard (event happens later)

Cox model in R

```
library(survival); library(broom); library(ggplot2)

# Kaplan-Meier estimator

km_fit = survfit(Surv(time, event) ~ group, data = d)

ggplot(tidy(km_fit), aes(time, estimate, color =
strata)) +

  geom_step() + geom_ribbon(aes(ymin = conf.low,
  ymax = conf.high, fill = strata), alpha = .2)

# Cox PH model

cox_fit = coxph(Surv(time, event) ~ x1 + x2, data = d)

summary(cox_fit) # shows hazard ratios

exp(coef(cox_fit))
```

Cox PH: R output

Using lung data (survival pkg): survival of lung cancer patients

```
> cox_fit = coxph(Surv(time, status) ~ age + sex + ph.ecog, data = lung)
> summary(cox_fit)
```

```
n= 227, number of events= 164
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	0.011067	1.011128	0.009267	1.194	0.23242
sex	-0.552612	0.575445	0.167739	-3.294	0.00099 ***
ph.ecog	0.463728	1.589991	0.113577	4.083	4.45e-05 ***

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0111	0.9890	0.9929	1.0297
sex	0.5754	1.7378	0.4142	0.7994
ph.ecog	1.5900	0.6289	1.2727	1.9864

```
Concordance= 0.637 (se = 0.025 )
```

```
Likelihood ratio test= 30.5 on 3 df, p=1e-06
```

```
Wald test = 29.93 on 3 df, p=1e-06
```

```
Score (logrank) test = 30.5 on 3 df, p=1e-06
```

- $\text{exp}(\text{coef}) = \text{hazard ratios}$ — the key quantity to report
- sex (2 = female): $\text{HR} = 0.58 \rightarrow 42\%$ *lower* hazard (longer survival)
- ph.ecog: $\text{HR} = 1.59 \rightarrow 59\%$ *higher* hazard per unit increase
- Concordance = 0.64: moderate predictive accuracy

Interpreting the Cox model

	Coef	Exp(coef)	SE	p-value
gdp_pc	-0.31	0.73	0.08	< 0.001
military	0.64	1.90	0.15	0.001
log_pop	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\text{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event

Interpreting the Cox model

	Coef	Exp(coef)	SE	p-value
gdp_pc	-0.31	0.73	0.08	< 0.001
military	0.64	1.90	0.15	0.001
log_pop	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\text{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event
- $\exp(\hat{\beta}_{\text{military}}) = 1.90$: military regimes face 90% **higher** hazard

Interpreting the Cox model

	Coef	Exp(coef)	SE	p-value
gdp_pc	-0.31	0.73	0.08	< 0.001
military	0.64	1.90	0.15	0.001
log_pop	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\text{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event
- $\exp(\hat{\beta}_{\text{military}}) = 1.90$: military regimes face 90% **higher** hazard
- Hazard ratio < 1: covariate *reduces* event rate $\rightarrow$ *increases* survival

Interpreting the Cox model

	Coef	Exp(coef)	SE	p-value
gdp_pc	-0.31	0.73	0.08	< 0.001
military	0.64	1.90	0.15	0.001
log_pop	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\text{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event
- $\exp(\hat{\beta}_{\text{military}}) = 1.90$: military regimes face 90% **higher** hazard
- Hazard ratio < 1: covariate *reduces* event rate $\rightarrow$ *increases* survival
- Hazard ratio > 1: covariate *increases* event rate $\rightarrow$ *decreases* survival

What if proportional hazards fails?

- Test with `cox.zph()`: significant p -value $\Rightarrow$ effect changes over time

What if proportional hazards fails?

- Test with `cox.zph()`: significant p -value $\Rightarrow$ effect changes over time
- **Solutions:**

What if proportional hazards fails?

- Test with `cox.zph()`: significant p -value $\Rightarrow$ effect changes over time
- **Solutions:**
 - **Stratify:** `coxph(Surv(t,d) ~ x + strata(z))`
Fits separate baseline hazard per group; useful for control variables

What if proportional hazards fails?

- Test with `cox.zph()`: significant p -value $\Rightarrow$ effect changes over time
- **Solutions:**
 - **Stratify:** `coxph(Surv(t,d) ~ x + strata(z))`
Fits separate baseline hazard per group; useful for control variables
 - **Time interaction:** let β vary with time
Models the changing effect explicitly

What if proportional hazards fails?

- Test with `cox.zph()`: significant p -value $\Rightarrow$ effect changes over time
- **Solutions:**
 - **Stratify:** `coxph(Surv(t,d) ~ x + strata(z))`
Fits separate baseline hazard per group; useful for control variables
 - **Time interaction:** let β vary with time
Models the changing effect explicitly
 - **Split follow-up:** divide time into periods, fit separate models
Simple but choice of cutpoint can be arbitrary

What if proportional hazards fails?

- Test with `cox.zph()`: significant p -value $\Rightarrow$ effect changes over time
- **Solutions:**
 - **Stratify:** `coxph(Surv(t,d) ~ x + strata(z))`
Fits separate baseline hazard per group; useful for control variables
 - **Time interaction:** let β vary with time
Models the changing effect explicitly
 - **Split follow-up:** divide time into periods, fit separate models
Simple but choice of cutpoint can be arbitrary
 - **Different model:** parametric (Weibull) or accelerated failure time
Do not require proportional hazards

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Pulling it together:

Which model for which data?

Choosing the right model

If your outcome is...	Use	Key check
Continuous	OLS	Linearity, homoskedasticity
Binary (0/1)	Logit / LPM	Rare events → prefer logit
Ordered categories	Ordered logit	Test PO assumption (Brant)
Unordered categories	Multinomial logit	IIA plausible?
Non-negative integer	Poisson	Test overdispersion
Overdispersed count	Neg. Binomial	Often the safe default
Many zeros	Zero-inflated	Two-process data?
Time until event	Cox PH	Test PH assumption

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R
- **Step 4:** Check assumptions

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R
- **Step 4:** Check assumptions
 - Counts: overdispersion test → switch to NegBin if needed

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R
- **Step 4:** Check assumptions
 - Counts: overdispersion test → switch to NegBin if needed
 - Ordinal: Brant test for proportional odds

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R
- **Step 4:** Check assumptions
 - Counts: overdispersion test → switch to NegBin if needed
 - Ordinal: Brant test for proportional odds
 - Nominal: IIA plausibility (substantive reasoning)

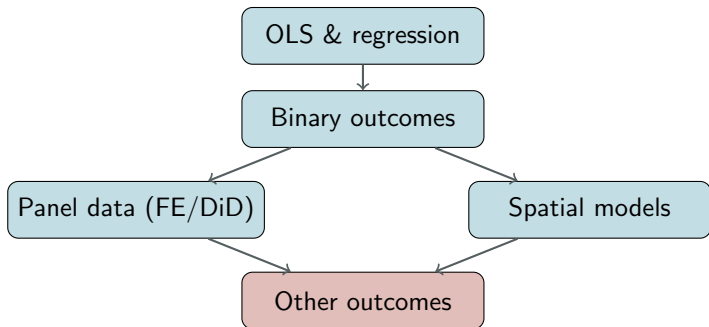
Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R
- **Step 4:** Check assumptions
 - Counts: overdispersion test → switch to NegBin if needed
 - Ordinal: Brant test for proportional odds
 - Nominal: IIA plausibility (substantive reasoning)
 - Duration: Schoenfeld residuals for PH assumption

Practical checklist

- **Step 1:** What is the DGP? Identify the outcome type
- **Step 2:** Choose model family accordingly (see table)
- **Step 3:** Estimate the model in R
- **Step 4:** Check assumptions
 - Counts: overdispersion test → switch to NegBin if needed
 - Ordinal: Brant test for proportional odds
 - Nominal: IIA plausibility (substantive reasoning)
 - Duration: Schoenfeld residuals for PH assumption
- **Steps 5–6:** Interpret via AMEs / predicted probabilities (**never raw coefficients alone**); visualize results

This semester



Today: ordinal, nominal, count, duration

(Computing session on April 23rd)

For the exam

- Be able to **identify** the right model given an outcome description

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)
 - Proportional hazards (Cox)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)
 - Proportional hazards (Cox)
- Interpret coefficients correctly:

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)
 - Proportional hazards (Cox)
- Interpret coefficients correctly:
 - Ordered/multinomial logit: AMEs or predicted probabilities

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)
 - Proportional hazards (Cox)
- Interpret coefficients correctly:
 - Ordered/multinomial logit: AMEs or predicted probabilities
 - Poisson/NegBin: $\exp(\hat{\beta}) = \text{incidence rate ratio}$

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)
 - Proportional hazards (Cox)
- Interpret coefficients correctly:
 - Ordered/multinomial logit: AMEs or predicted probabilities
 - Poisson/NegBin: $\exp(\hat{\beta}) = \text{incidence rate ratio}$
 - Cox PH: $\exp(\hat{\beta}) = \text{hazard ratio}$

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
 - Proportional odds (ordered logit)
 - IIA (multinomial logit)
 - Equidispersion (Poisson)
 - Proportional hazards (Cox)
- Interpret coefficients correctly:
 - Ordered/multinomial logit: AMEs or predicted probabilities
 - Poisson/NegBin: $\exp(\hat{\beta}) = \text{incidence rate ratio}$
 - Cox PH: $\exp(\hat{\beta}) = \text{hazard ratio}$
- Know the relevant **R functions**

Questions?